436: A Comparison of Participation in Intercollegiate Athletics

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Introduction

The data for this task come from Equity in Athletics Data Analysis.

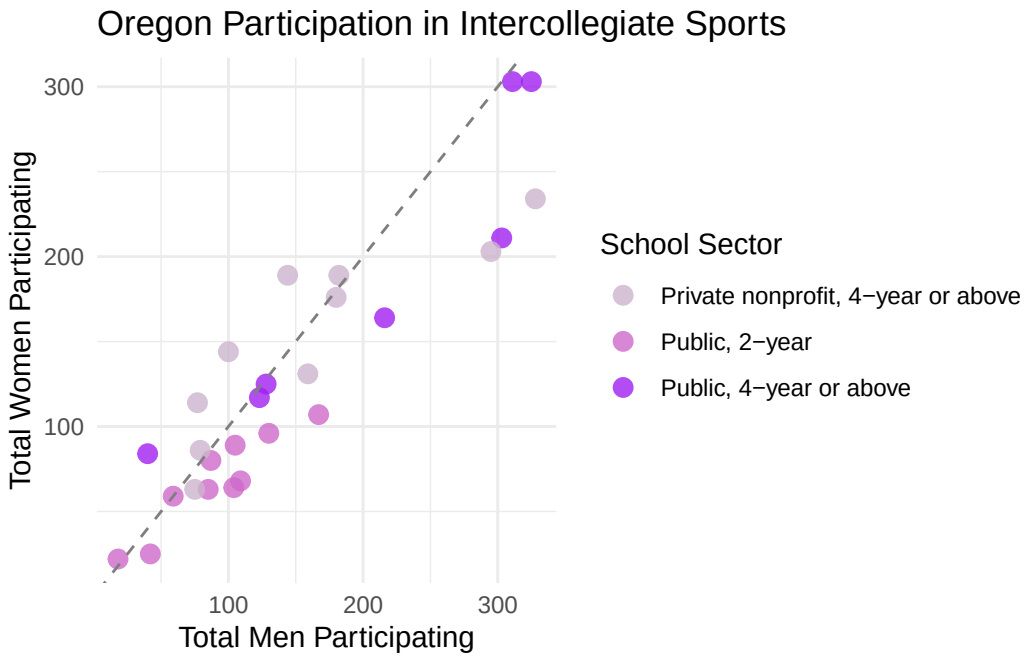
"This database consists of athletics data that are submitted annually as required by the Equity in Athletics Disclosure Act (EADA), via a Web-based data collection, by all co-educational postsecondary institutions that receive Title IV funding (i.e., those that participate in federal student aid programs) and that have an intercollegiate athletics program."

The data will be used to answer these two questions of interest:

- For schools in Oregon, are there differences in the number of men and women participating in intercollegiate sports across the different types of school?
- Which schools in Oregon have the highest level of participation in intercollegiate sports?

Question 1: For schools in Oregon, are there differences in the number of men and women participating in intercollegiate sports across the different types of school?

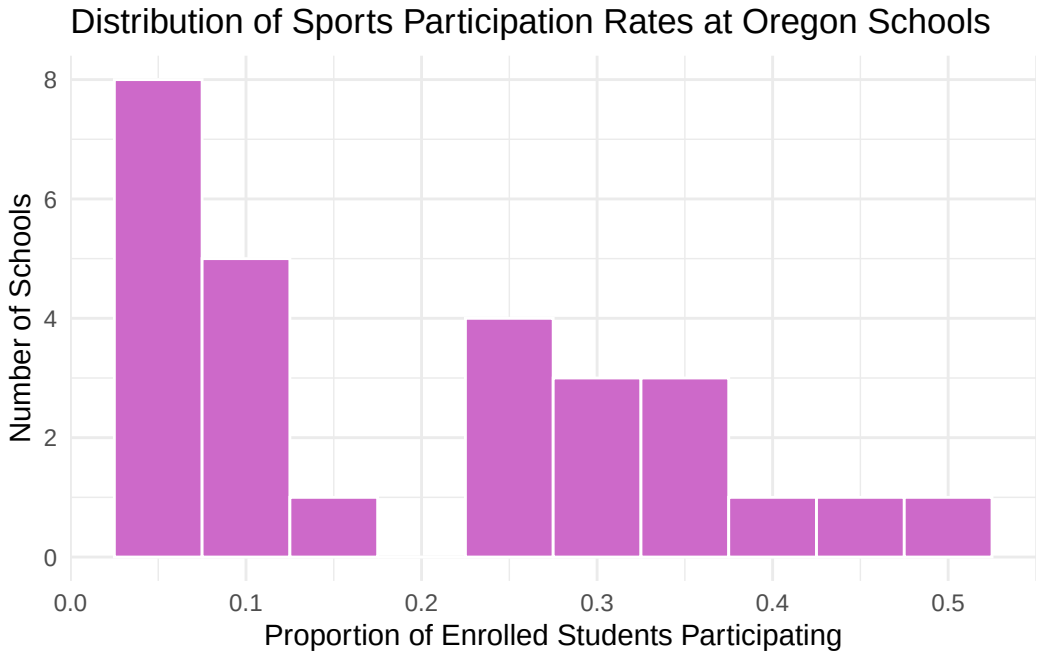
To explore this question here is a scatterplot of Oregon women vs. men participation in intercollegiate athletics colored by sector



School sector does relate to participation differences. Public 2-year schools consistently skew toward higher male participation, while private 4-year schools exhibit much more variation in gender balance. Public 4-year schools have the highest total participation but also tend to lean toward equal or higher male participation. Across all three sectors do we see a majority have more male participation.

Question 2: Which schools in Oregon have the highest level of participation in intercollegiate sports?

To start answering this question this histogram is built to showcase the distribution of sports participation



From this histogram it shows that the largest group of institutions has a relatively low proportion of athletes. Specifically, 14 schools have less than 15% (0.15) of their student body participating in sports. There is a second, widespread group of schools on the right side of the histogram where athletics make up a massive portion of the student population. As can be see it ranges from about 25% all the way up to over 50%

Now this table is constructed to showcase the schools with over 40% participation to highlight the very top end

Table 1: Oregon Schools with the Highest Sports Participation Rates

institution	sector	total_partic	total_enroll	participate
Multnomah University	Private nonprofit, 4-year or above	138	281	0.491
Bushnell University	Private nonprofit, 4-year or above	191	423	0.452
Corban University	Private nonprofit, 4-year or above	333	793	0.420

From this table we can answer that the schools in Oregon with the highest levels of participation in intercollegiate sports are Multnomah University, Bushnell University, and Corban University.

All three of these top institutions fall under the "Private nonprofit, 4-year or above" sector. Their high participation rates are driven by their relatively small total enrollments (all under 800 students), meaning that even a realtively small amount of student-athletes can make up nearly half of the entire student body. As is the case with these schools.